

IT Project Management

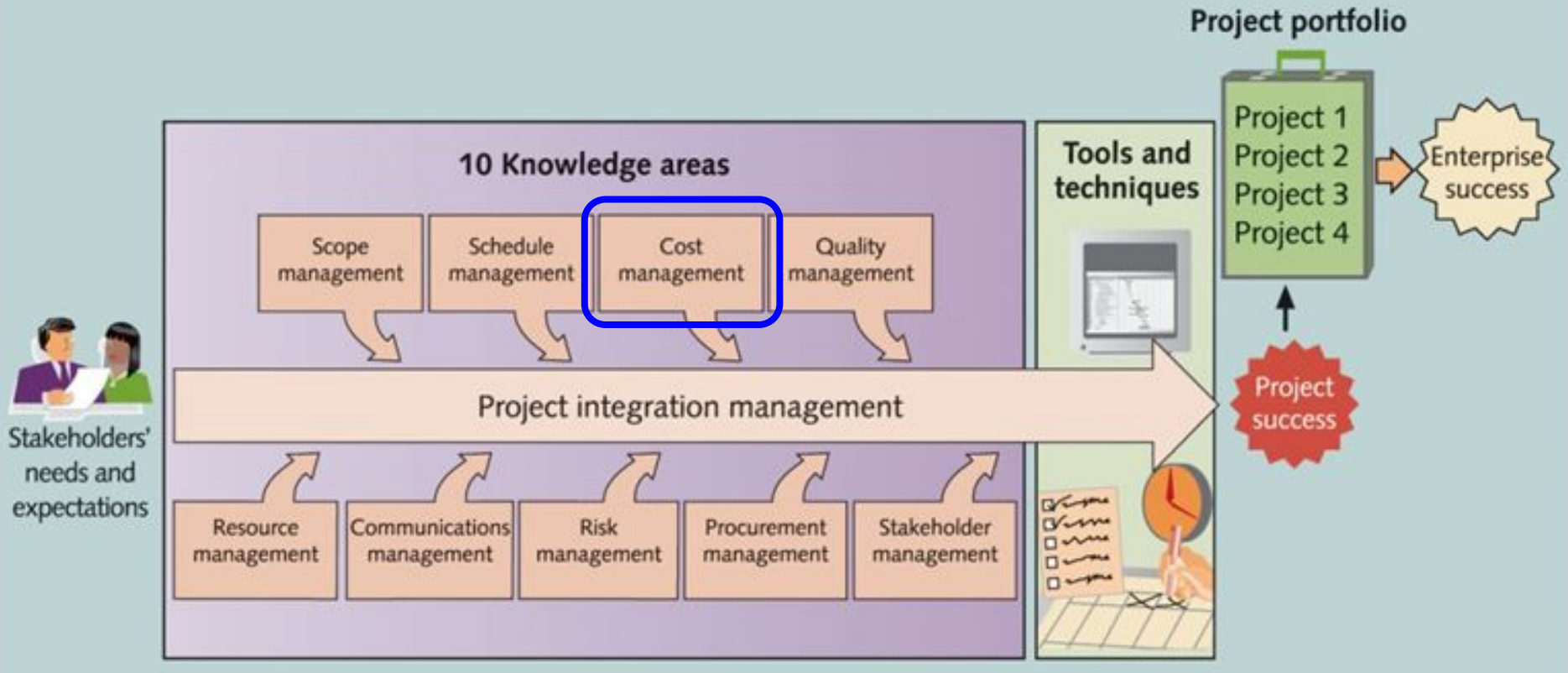
Lecture 6: Project **Cost** Management

—*Budgeting, Tracking, and Controlling Expenses*—

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Project Cost Management



About this lecture



Learning objectives:

- **Understand key concepts** of project cost management
- **Explain cost estimation techniques** and their applications
- **Apply Earned Value Management (EVM)** for project cost tracking
- **Identify risks and cost control methods** to ensure project success

Introduction to Project Cost Management

What is Project Cost?

Cost is the **sacrifice or expenditure of resources** to achieve a **project goal**

Key Cost Elements:

- **Labor Wages:** Paying project staff
- **Equipment & Tools:** Buying or renting machines, computers, etc.
- **Materials & Supplies:** Raw materials, office supplies
- **Travel & Logistics:** Employee transportation costs
- **Software & Licensing:** IT and software project costs
- **Training Costs:** Staff education and skill development
- **Saving Time:** Adding resources to compression task duration

What is Project Cost Management?

The process of **planning, estimating, budgeting, and controlling** costs

→ To ensure a project **is completed within the approved budget**

→ Includes **tracking financial performance** and making adjustments

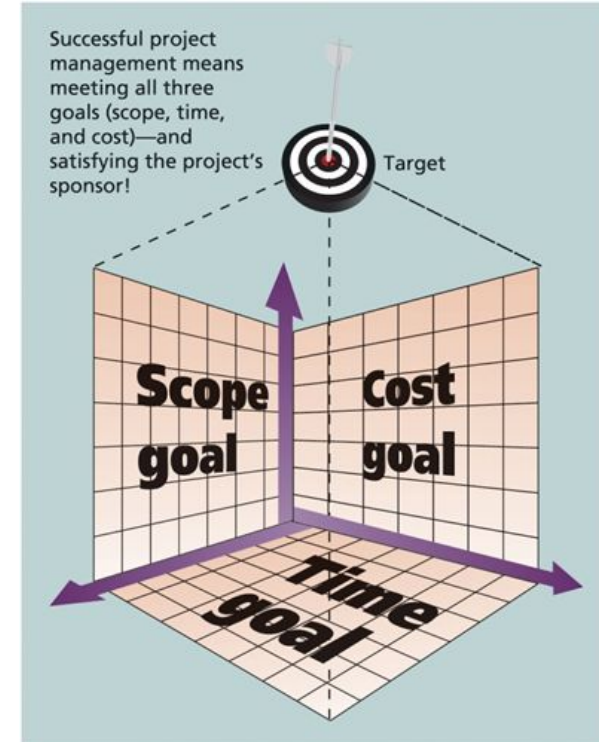
The Triple Constraint: Scope, Time, Cost

- **Scope:** What is included in the project?
- **Time:** How long will it take?
- **Cost:** How much will it cost?

Increasing scope → **Higher cost & longer time**

Reducing time → **Higher cost to speed up work**

Lowering cost → **May reduce scope or extend timeline**



Project Cost Failures in IT

- **1995 CHAOS Report:** Average project cost overrun was **189%**
 - **Global IT Project lost \$81 billion** due to poor financial tracking
- **2010 Study:** Improved to **43%** but still a major issue
 - The **absolute loss value is high** because total budget increased

Project Cost Failures in general

Sydney Opera House	Planned at \$7M	Completed at \$102M (1400% over budget)	Reason: poor initial estimates
Metro Line 1 (Bến Thành - Suối Tiên)	Planned at \$1.09B	Increased to \$2.05B (also increased scope and time)	Reason: design changes and funding delays

Project Cost Failures: Common Causes

- Poor initial cost estimates
- Scope creep (unplanned changes)
- Technical & design challenges
- Weak cost monitoring and control

→ Cost management **provides tools and techniques** to address the causes

Project Cost Management Processes

The 4 Key Steps of Project Cost Management Process

1. **Plan Cost Management:**

Defines how costs will be **estimated, budgeted, and tracked**

2. **Estimate Costs:**

Identifies costs for **labor, materials, and resources**

3. **Determine Budget:**

Allocates the **total cost** across project phases

4. **Control Costs:**

Monitors **actual vs. planned costs** to adjust budgets if needed

Step 1: Plan Cost Management

What is Cost Management Planning?

- The process of defining **how** project costs will be **estimated, budgeted, managed, monitored, and controlled**
- Ensures **consistency and accuracy** in financial planning

→ It is important to define how costs will be managed **before a project starts**

Step 1: Plan Cost Management

Key Elements of a Cost Management Plan:

- **Measurement Units:** Define cost in **USD, labor hours, materials, etc.**
- **Cost Baseline:** Approved version of the budget for performance tracking
- **Control Thresholds:** Defines acceptable **cost variations** (e.g., $\pm 10\%$)
- **Reporting Frequency:** Weekly/monthly cost reports for tracking
- **Funding Limits:** Ensuring cash flow aligns with project needs
- **Process for Adjusting Costs:** How cost overruns will be handled

Step 2: Estimate Costs

What is Cost Estimation?

- Determines **the cost of resources, labor, equipment, and services** needed for the project, includes **direct and indirect costs**
- Helps avoid **under-budgeting or over-budgeting**

Step 2: Estimate Costs

Cost Estimation Methods:

- **Top-down Estimating:**
 - Based on **past similar projects**
- **Bottom-up Estimating:**
 - Calculate for **each component** and summed up
- **Parametric Estimating:**
 - Uses **historical data & mathematical models** (e.g., cost per unit)
- **Three-Point Estimating (PERT):**
 - **Optimistic (O), Most Likely (M), and Pessimistic (P)** estimates
- **Monte Carlo Simulation:**
 - Uses **probability-based forecasting**

Step 3: Determine Budget

What is Budget Determination?

- Combines **all estimated costs** to create a **total project budget**
- Ensures costs are **allocated properly across project phases**
- Establishes a **Cost Baseline** for monitoring financial performance

Step 3: Determine Budget

Components of a Project Budget:

- **Cost Aggregation:** Summing up all cost estimates
- **Contingency Reserves:** Extra funds for **known risks**
- **Management Reserves:** Extra funds for **unknown risks**
- **Funding Requirements:** Ensuring the project has **available cash flow**

→ Projects need both **reserves** for both **known risks** and **unknown risks**

Step 4: Control Costs

What is Cost Control?

- Ensures project **spending stays within the budget**
- Helps identify **overspending early** and take corrective action
- Uses **Earned Value Management (EVM)** to track financial performance
 - Improves **decision-making for cost adjustments**

Cost Estimation Techniques

Top-Down Estimating (Analogous Estimation)

- Uses **historical cost data** from similar past projects to estimate the current project's cost
- **Quick and simple**, but **less accurate**

When to Use:

- Early project planning (before detailed data is available)
- When a company has **historical cost data from similar projects**
- Best for **high-level cost estimates**

Top-Down Estimating (Analogous Estimation)

Example:

- A **software company** estimates the cost of a new **mobile app** based on a **previous similar app development**
- A **construction firm** estimates a **new office building cost** based on a **previously built office of similar size**

Bottom-Up Estimating

- Breaks the project down into **small components**, then estimates **each component separately**
- **More accurate method**, but also **time-consuming**

When to Use:

- When **detailed cost breakdowns** are required
- For **complex projects** where accuracy is critical
- Used in **IT, construction, and large-scale engineering projects**

Bottom-Up Estimating

Process:

- Break down work into **Work Breakdown Structure (WBS)**
- Estimate **each component cost** separately
- Add up all the estimated costs to get the **total project cost**

Example: Webapp Development Project with

- Frontend: \$3,000
 - Backend: \$5,000
 - Database: \$4,000
- Total Cost: \$12,000

Parametric Estimating

- Uses **mathematical formulas and cost per unit data** to estimate project costs
- **More accurate than top-down (analogous) estimating**

$$\text{Estimated Cost} = \text{Cost per Unit} \times \text{Number of Units}$$

When to Use:

- When **cost per unit** (e.g., per square meter, per software feature) is known
- When projects are **repeatable and scalable**
- Works well for **construction, manufacturing, and software projects**

Three-Point Estimating (PERT)

Uses **three estimates** to predict costs:

- **Optimistic (O):** Best-case scenario
- **Most Likely (M):** Expected scenario
- **Pessimistic (P):** Worst-case scenario

→ We have learned this in previous lecture

$$E = \frac{O + 4M + P}{6}$$

Monte Carlo Simulation

- A **probability-based** estimation method
- Runs **thousands of simulations**
- Provide more **accurate and detailed estimation** of project cost

→ We have learned this in previous lecture

Estimate Costs: Example (Bottom-up and Parametric)

	# Units/Hrs.	Cost/Unit/Hr.	Subtotals	WBS Level 2 Totals	% of Total
WBS Items					
1. Project Management				\$306,300	20%
Project manager	960	\$100	\$96,000		
Project team members	1920	\$75	\$144,000		
Contractors (10% of software development and testing)			\$66,300		
2. Hardware				\$76,000	5%
2.1 Handheld devices	100	\$600	\$60,000		
2.2 Servers	4	\$4,000	\$16,000		
3. Software				\$614,000	40%
3.1 Licensed software	100	\$200	\$20,000		
3.2 Software development*			\$594,000		
4. Testing (10% of total hardware and software costs)			\$69,000	\$69,000	5%
5. Training and Support				\$202,400	13%
Trainee cost	100	\$500	\$50,000		
Travel cost	12	\$700	\$8,400		
Project team members	1920	\$75	\$144,000		
6. Reserves (20% of total estimate)			\$253,540	\$253,540	17%
Total project cost estimate				\$1,521,240	

Budget Determination

Creating a Budget by Cost Aggregation

Cost Aggregation:

- **Adds up individual cost estimates into higher-level budgets**
- **Uses the Work Breakdown Structure (WBS) to summarize total costs**

Creating a Budget by Cost Aggregation

Steps in Cost Aggregation to ensure no cost elements are missed:

1. **Estimate Costs for Each Task** (e.g., labor, materials, overhead)
2. **Group Costs into Work Packages** (subproject or department level)
3. **Summarize Work Packages into the Total Project Budget**

Handling Risk with Budget Reserves

No project runs exactly as planned: **unexpected costs always arise**

→ Reserves help absorb **cost overruns, risks, and scope changes**

Type	Purpose	Who Approves It?	Example
Contingency Reserves	Covers known risks identified in risk management	Project Manager	Unexpected supplier delays
Management Reserves	Covers unknown risks or unforeseen scope changes	Senior Management	Economic inflation, new regulations

Funding Requirements: Secure Sufficient Project Funding

- Defines **how much money is needed** at different phases of the project
- Ensures **continuous cash flow** so the project doesn't stall
- Helps stakeholders plan for **financing, loans, or investment rounds**

Funding Requirements: Secure Sufficient Project Funding

Key Considerations:

- **Total Budget Needed** – Includes all costs and reserves
- **Spending Timeline** – When are funds required (monthly, quarterly, yearly)?
- **Source of Funding** – Internal budget, bank loans, investor capital, grants, etc.

Example:

- **A Startup Project:** Raises funding in **three rounds** (Seed, Series A, Series B)
- **A Government Project:** Receives **funding in stages** based on project progress

Determine Budget: Example (Cost Baseline)

WBS Items	1	2	3	4	5	6	7	8	9	10	11	12	Totals
1. Project Management													
1.1 Project manager	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	96,000
1.2 Project team members	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	144,000
1.3 Contractors		6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	66,300
2. Hardware													
2.1 Handheld devices				30,000	30,000								60,000
2.2 Servers				8,000	8,000								16,000
3. Software													
3.1 Licensed software				10,000	10,000								20,000
3.2 Software development		60,000	60,000	80,000	127,000	127,000	90,000	50,000					594,000
4. Testing			6,000	8,000	12,000	15,000	15,000	13,000					69,000
5. Training and Support													
5.1 Trainee cost									50,000				50,000
5.2 Travel cost									8,400				8,400
5.3 Project team members							24,000	24,000	24,000	24,000	24,000	24,000	144,000
6. Reserves				10,000	10,000	30,000	30,000	60,000	40,000	40,000	30,000	3,540	253,540
Totals	20,000	86,027	92,027	172,027	223,027	198,027	185,027	173,027	148,427	90,027	80,027	53,567	1,521,240

Cost Control & Monitoring

Variance Analysis

- Compares **Planned Costs vs. Actual Costs** to find **deviations**
- Helps determine whether the project is **on track or over budget**

Variance Analysis

Key Metrics:

- **Planned Value (PV):** Budgeted cost of **planned work percentage**
- **Earned Value (EV):** Budgeted cost of **completed work percentage**
- **Actual Cost (AC):** The actual money spent so far
- **Cost Variance (CV) = $EV - AC$:** Measures **budget performance**
- **Cost Performance Index (CPI) = EV / AC :** Indicates **cost efficiency**

→ Provide some simple monitoring tools

→ Use EVM for more advanced monitoring

Earned Value Management (EVM)

What is EVM?

- A method to **track cost, schedule, and performance** against the baseline
- Helps **identify deviations early** and take corrective actions
- Combines **scope, time, and cost** to assess project health
 - Measures project progress objectively
 - Helps forecast cost overruns or schedule delays
 - Provides a single system for cost & schedule tracking

Earned Value Management (EVM): Key Metrics & Formulas

1. **Planned Value (PV):** Budgeted cost of work **scheduled** at a given point

Formula: $PV = (\text{Planned \% of Work}) \times (\text{Total Budget})$

2. **Earned Value (EV):** Budgeted cost of work **actually completed** at a given point

Formula: $EV = (\text{Actual \% of Work Completed}) \times (\text{Total Budget})$

3. **Actual Cost (AC):** **Actual** money spent on the work completed so far

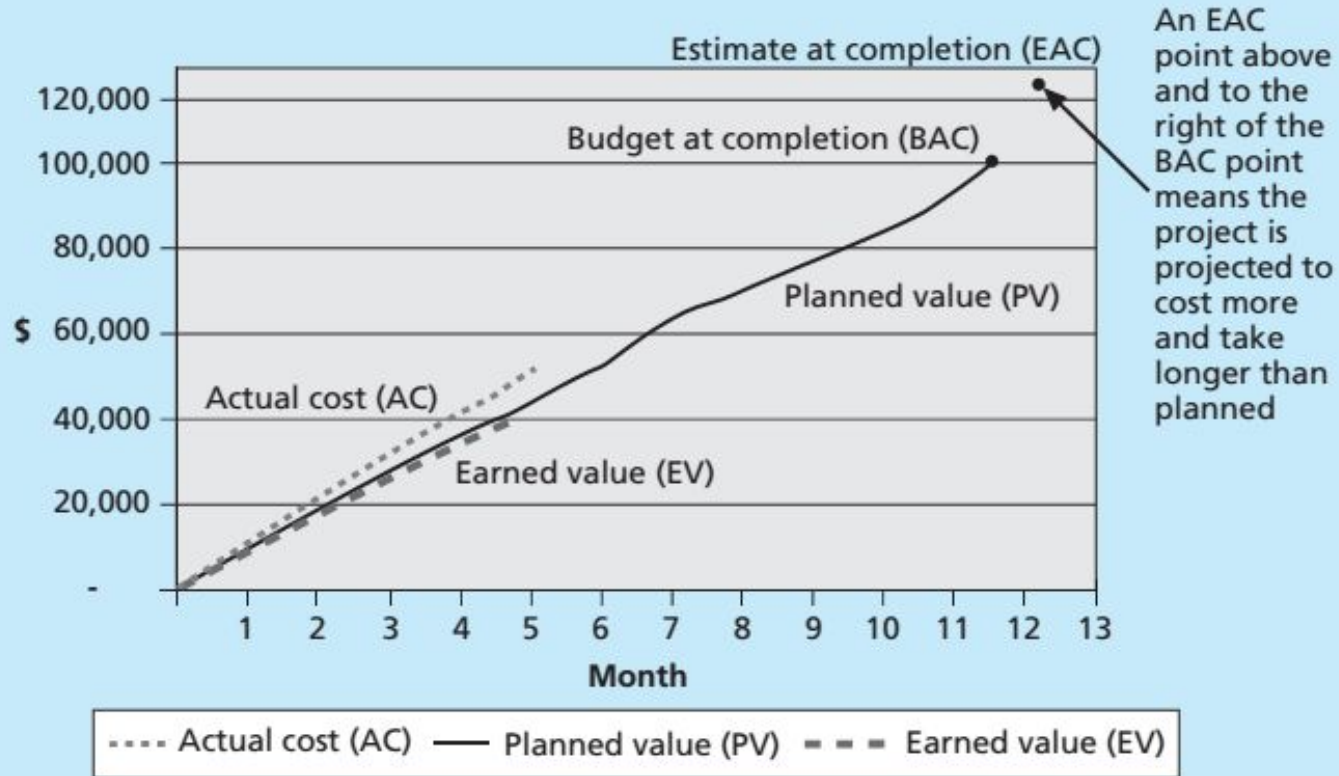
Earned Value Management (EVM): Key Metrics & Formulas

Metric	Formula	Interpretation
Cost Variance (CV)	$CV = EV - AC$	(+) Under Budget, (-) Over Budget
Schedule Variance (SV)	$SV = EV - PV$	(+) Ahead of Schedule, (-) Behind Schedule
Cost Performance Index (CPI)	$CPI = EV / AC$	$CPI > 1$ = Under Budget, $CPI < 1$ = Over Budget
Schedule Performance Index (SPI)	$SPI = EV / PV$	$SPI > 1$ = Ahead of Schedule, $SPI < 1$ = Delayed
To-Complete Performance Index (TCPI)	$TCPI = (BAC - EV) / (BAC - AC)$	> 1 = tough recovery, < 1 = easier recovery

Other Metrics:

- Initial project Total Budget: **BAC** (Budget At Completion)
- Initial project Total Time: **TAC** (Time At Completion)
- Re-estimate Total Amount: **EAC** (Estimate At Completion)
 - For Budget: **EAC_budget** = BAC / CPI
 - For Time: **EAC_time** = TAC / SPI
- Re-estimate Remaining Amount: **ETC** (Estimate To Complete)
 - For Budget: **ETC_budget** = $(BAC - EV) / CPI = EAC_budget - AC$
 - For Time: **ETC_time** = $EAC_time - Current_time$

Earned Value Management (EVM): Example



Earned Value Management (EVM): Example

Example Calculation:

- **Project Budget:** \$100,000
- **Planned Work Completion:** 50%
- **Actual Work Completed:** 40%
- **Actual Cost Spent:** \$45,000

Earned Value Management (EVM): Example

Example Calculation:

- **Project Budget:** \$100,000
- **Planned Work Completion:** 50%
- **Actual Work Completed:** 40%
- **Actual Cost Spent:** \$45,000

$$PV = 50\% \times 100,000 = 50,000$$

$$EV = 40\% \times 100,000 = 40,000$$

$$AC = 45,000$$

Earned Value Management (EVM): Example

Schedule Variance (SV):

$$SV = 40,000 - 50,000 = -10,000 \quad (\text{Behind Schedule})$$

Cost Variance (CV):

$$CV = 40,000 - 45,000 = -5,000 \quad (\text{Over Budget})$$

Schedule Performance Index (SPI):

$$SPI = \frac{40,000}{50,000} = 0.8 \quad (\text{Project is Behind Schedule})$$

Cost Performance Index (CPI):

$$CPI = \frac{40,000}{45,000} = 0.89 \quad (\text{Over Budget})$$

Recover from Cost Overruns

Using TCPI (To-Complete Performance Index):

- Shows how to **efficiently spend remaining funds** to stay within budget
- Helps project managers **decide whether cost recovery is possible**

Recover from Cost Overruns

Interpreting TCPI:

- **TCPI > 1.2** → Very difficult recovery (Need to reevaluate budget)
- **TCPI ~ 1.0 – 1.2** → Challenging but possible (Need to tighten cost controls)
- **TCPI < 1.0** → Easily on-track (Have managed funds wisely)

From previous example:

Budget you would need
for the remaining work

Budget you actually have
for the remaining work


$$\text{TCPI} = (\text{BAC} - \text{EV}) / (\text{BAC} - \text{AC})$$

$$= (100,000 - 40,000) / (100,000 - 45,000) = 1.09 \quad (\text{Challenging})$$

Recover from Cost Overruns

Recovery Strategies:

- **Fast-Tracking:** Work on some tasks **in parallel** to save time
- **Crashing:** Allocate **extra resources** (but increases costs)
- **Scope Adjustment:** Reduce non-critical features to cut costs
- **Renegotiate Vendor Contracts:** Lower supplier costs

Exercise

Exercise 1

A software company is building a **new mobile app** with a budget of **\$200,000**.

- **Planned Value (PV):** \$80,000
- **Earned Value (EV):** \$60,000
- **Actual Cost (AC):** \$75,000

Questions:

1. Calculate **Cost Variance (CV)** and **CPI** – Is the project (budget) on track?
2. Compute **Schedule Variance (SV)** and **SPI** – Is the project (schedule) on track?
3. Find **TCPI** – Can the project recover?

$$CV = EV - AC$$

$$CPI = EV / AC$$

$$SV = EV - PV$$

$$SPI = EV / PV$$

$$TCPI = (BAC - EV) / (BAC - AC)$$

BAC (Budget At Completion)

TAC (Time At Completion)

$$EAC_budget = BAC / CPI$$

$$EAC_time = TAC / SPI$$

$$ETC_budget = EAC_budget - AC$$

$$ETC_time = EAC_time - Current_time$$

Exercise 2

Given the following project information for a 1-year project:

- **Planned Value (PV)** = \$23,000
- **Earned Value (EV)** = \$20,000
- **Actual Cost (AC)** = \$25,000
- **Budget at Completion (BAC)** = \$120,000

Questions:

1. Calculate **Cost Variance (CV)** and **CPI**. Is the project **over budget**?
2. Calculate **Schedule Variance (SV)** and **SPI**. Will the project be completed **behind or ahead of schedule**?
3. Recalculate the new budget (**EAC**) for this project.
4. Reestimate how long it will take (**ETC**) to complete the project.

$$CV = EV - AC$$

$$CPI = EV / AC$$

$$SV = EV - PV$$

$$SPI = EV / PV$$

$$TCPI = (BAC - EV) / (BAC - AC)$$

BAC (Budget At Completion)

TAC (Time At Completion)

$$EAC_budget = BAC / CPI$$

$$EAC_time = TAC / SPI$$

$$ETC_budget = EAC_budget - AC$$

$$ETC_time = EAC_time - Current_time$$

Summary

Summary



You have learned to:

- **Understand key concepts** of project cost management
- **Explain cost estimation techniques** and their applications
- **Apply Earned Value Management (EVM)** for project cost tracking
- **Identify risks and cost control methods** to ensure project success

Thank you for listening

Q & A